SYSTEM INTEGRATION PRACTICES

Integrating Services

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Objective

- Identify some of the common services of an integrated system
- Understand the issues that can result from integrating systems that have these services

Integrated System Services Overview

- Modern integrated systems rely on various types of services, including:
  - **Naming Services** – Identifying and locating services.
  - **Security Services** – Data security, authentication, and access control.
  - **Authentication & Access Control** – Identity verification and access control.
  - **Reliability Services** – Ensuring system reliability and failure recovery.
  - **Load Balancing** – Distributing workloads to ensure system stability.
- Each of these services is essential in data integration platforms such as **SQL Server Integration Services (SSIS), Azure Data Factory (ADF), Talend, Apache NiFi, MuleSoft Anypoint Platform**.

Naming & Naming

Concept

- Naming Service is a service that maps names to actual addresses of services or resources.
- Common example: **DNS (Domain Name System)** maps domain names to IP addresses.

Real-world Applications

  - **SSIS & ADF**: Use **Connection Managers** to name and manage data sources.
  - **Apache NiFi**: Uses **FlowFile Attributes** to identify processed data.
  - **MuleSoft**: Uses **Service Registry** to manage API endpoints.

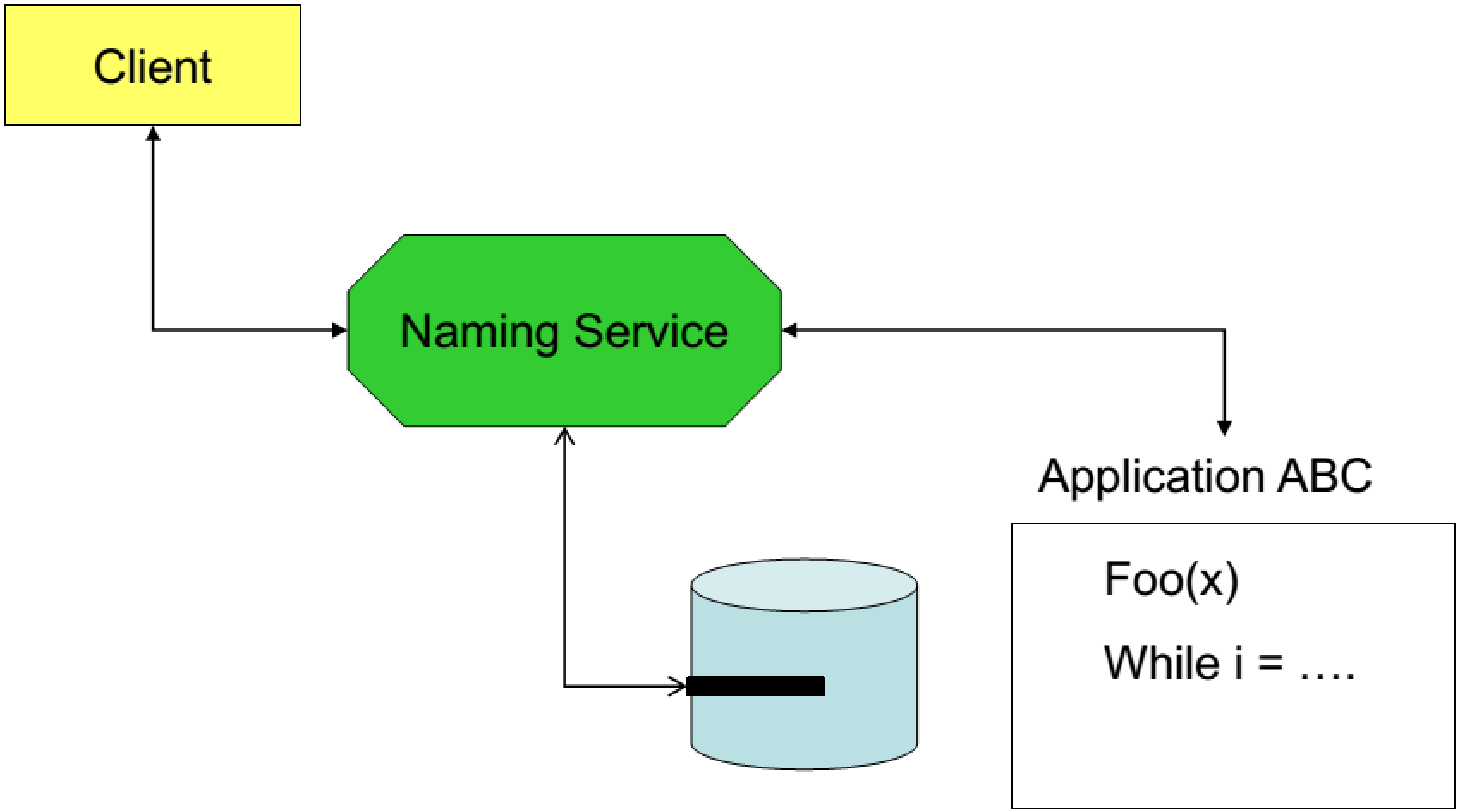
Challenges:

- Name conflicts when integrating multiple systems.
- Systems may enter an inconsistent state due to name changes.

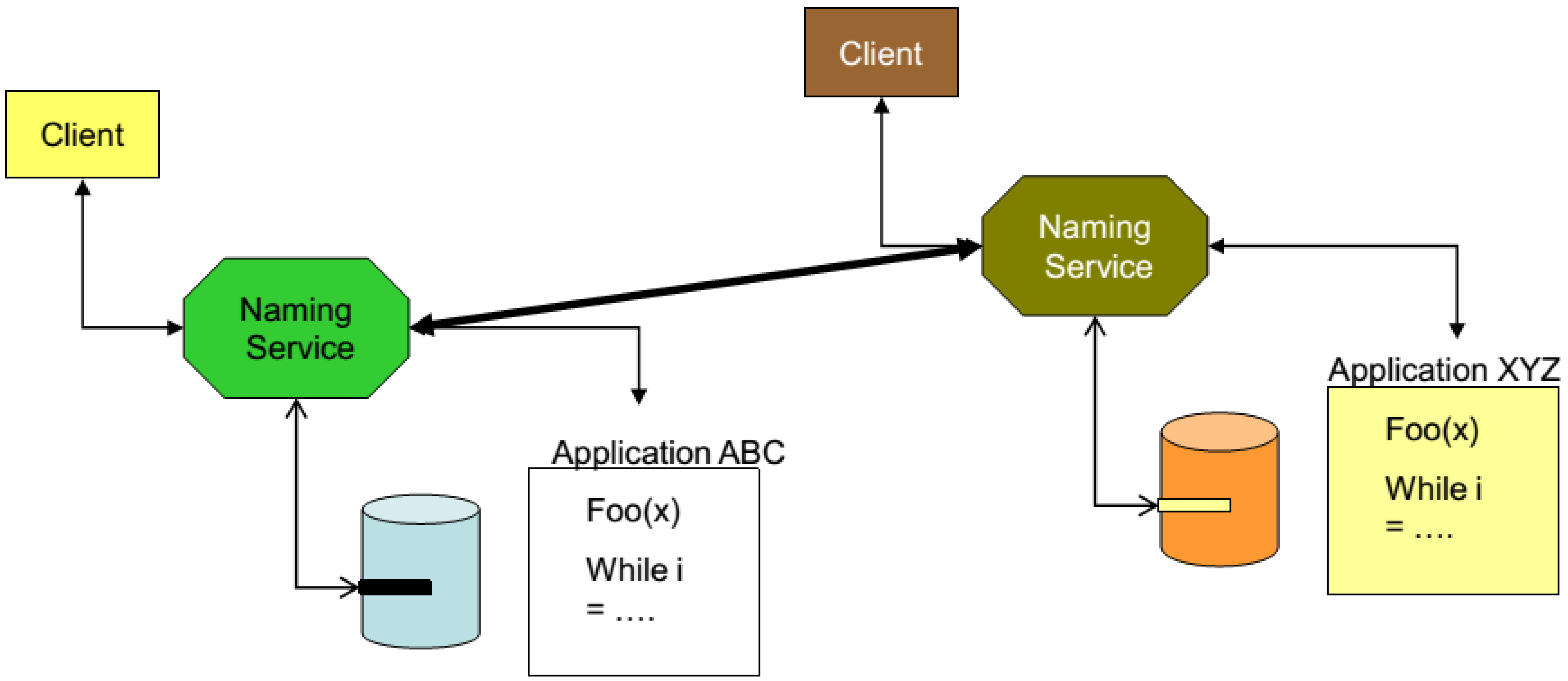
Integrated System Services: Example

192.31.7.130	CISCO.COM
204.71.177.35	YAHOO.COM
152.163.210.7	AOL.COM
198.150.15.234	MAT-MADISON.COM
207.46.131.15	MICROSOFT.COM
192.233.80.9	NOVELL.COM

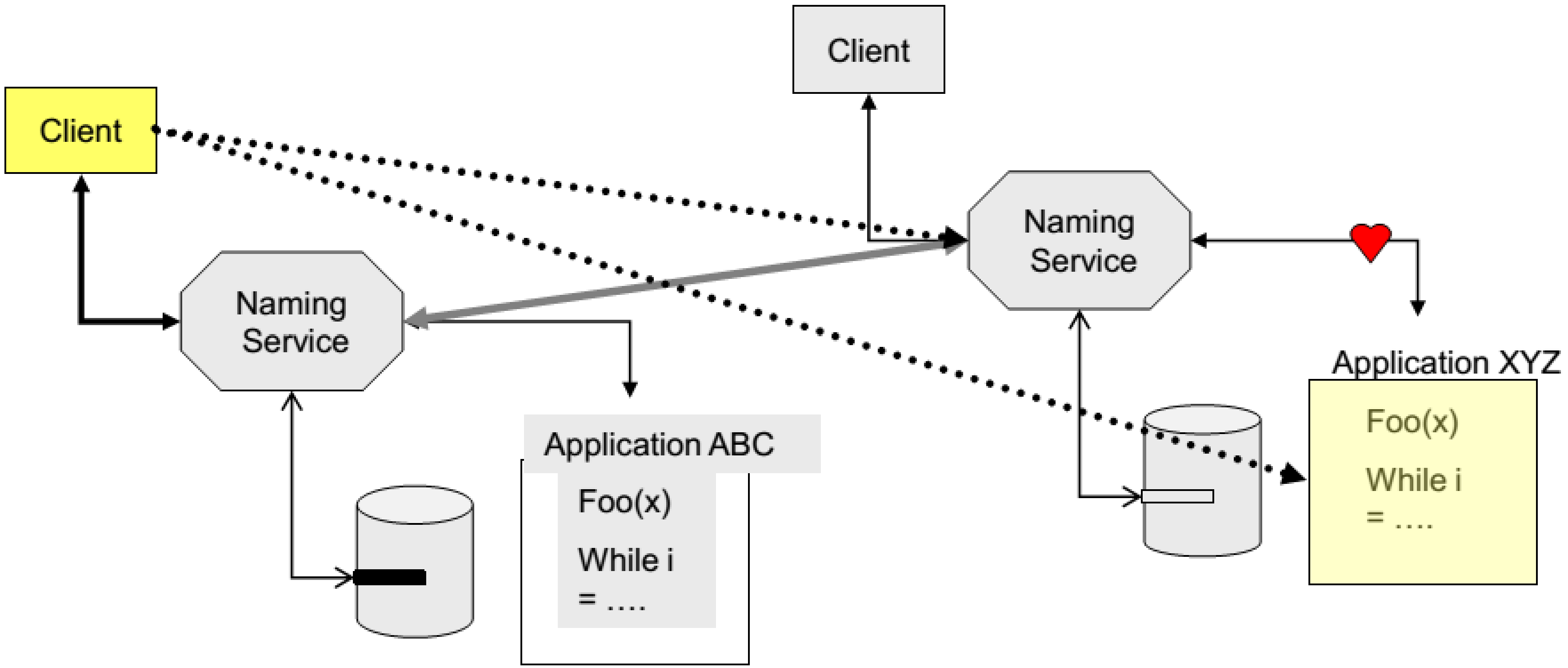
Integrated System Services: Naming Services



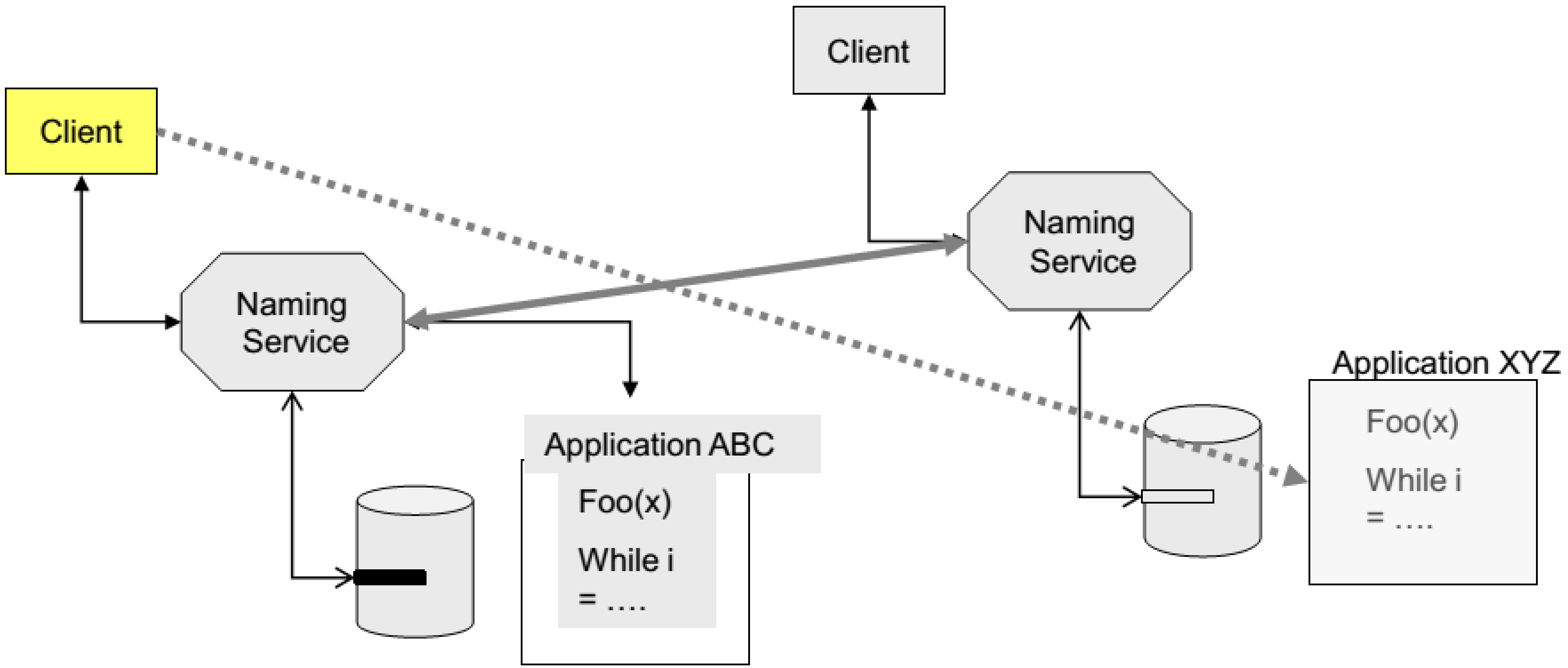
Integrated System Services: Naming Services



Naming Services



Naming Services: Inconsistent State



Global Services of an Integrated: Security

A **Global Integrated System** requires **Security Services** to ensure safe access and usage of resources.

Principles of Security Services:

1. Authentication

- All entities in the system must be **authenticated** to verify their identity.
- This can be done through passwords, tokens, digital certificates (SSL/TLS), or two-factor authentication (2FA).

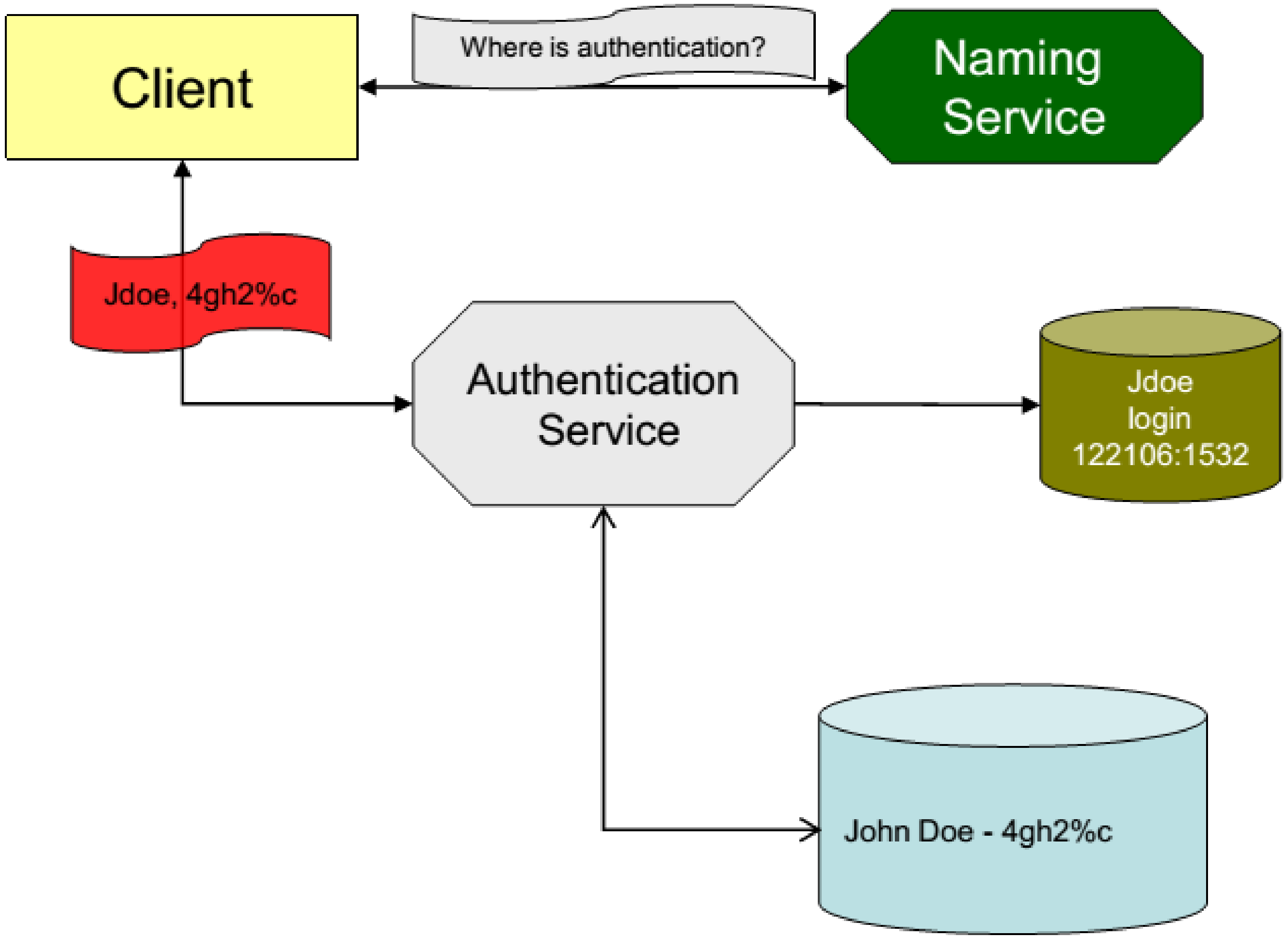
2. Access Control Management

- The system uses **access policies** to determine who can access which resources.
- Models such as **Role-Based Access Control (RBAC)** or **Attribute-Based Access Control (ABAC)** can be applied.
- Example: **Regular users** can only view data. **Administrators** can create, modify, and delete data.

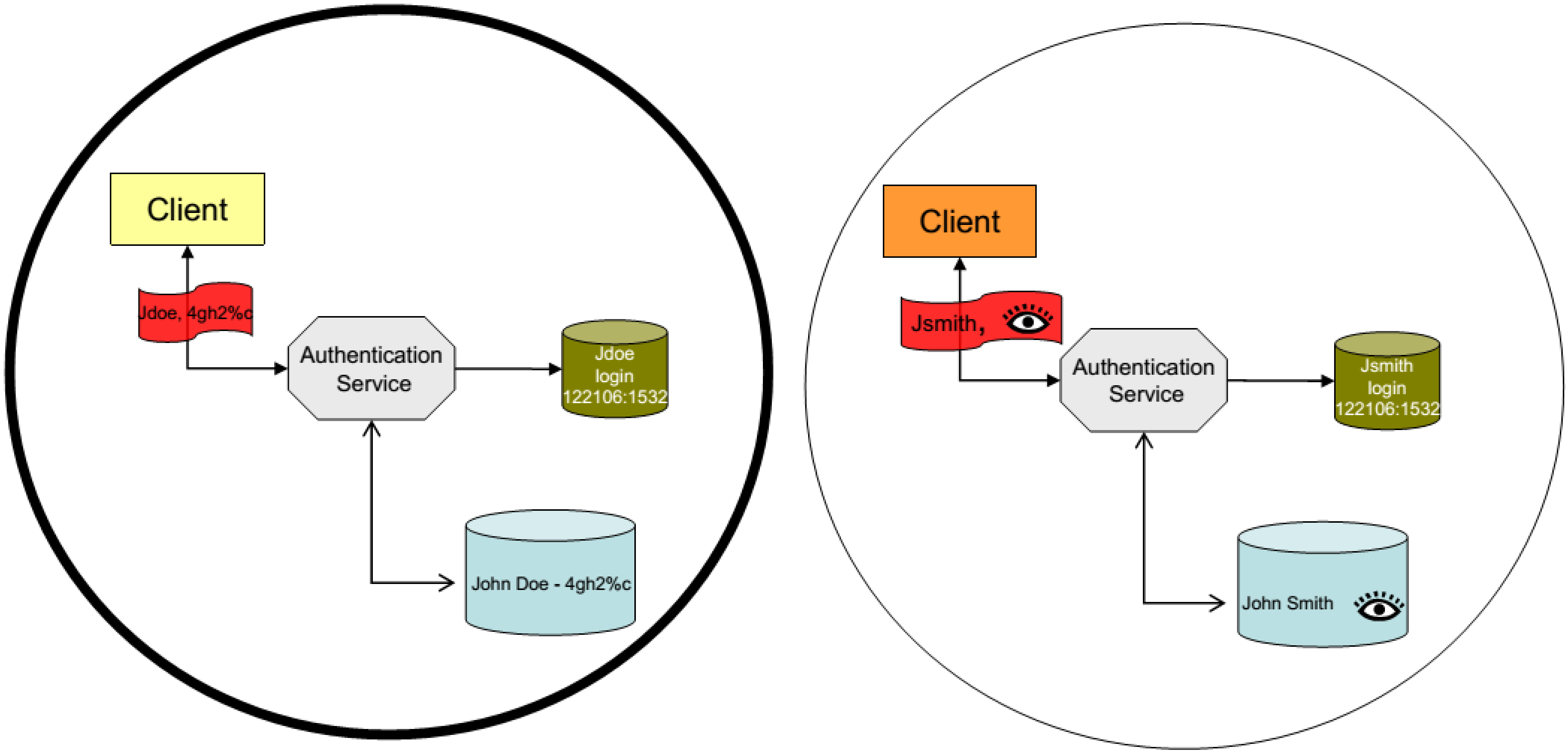
3. Activity Logging (Logging & Accountability)

1. All activities in the system can be **logged** for monitoring and auditing.
2. Helps detect anomalies, handle security incidents, and recover from failures.
3. **Different systems log different events**, for example:
    1. Network systems log connection data.
    2. Application systems log login attempts and file access.
    3. Databases log query executions and data modifications.

Security Services



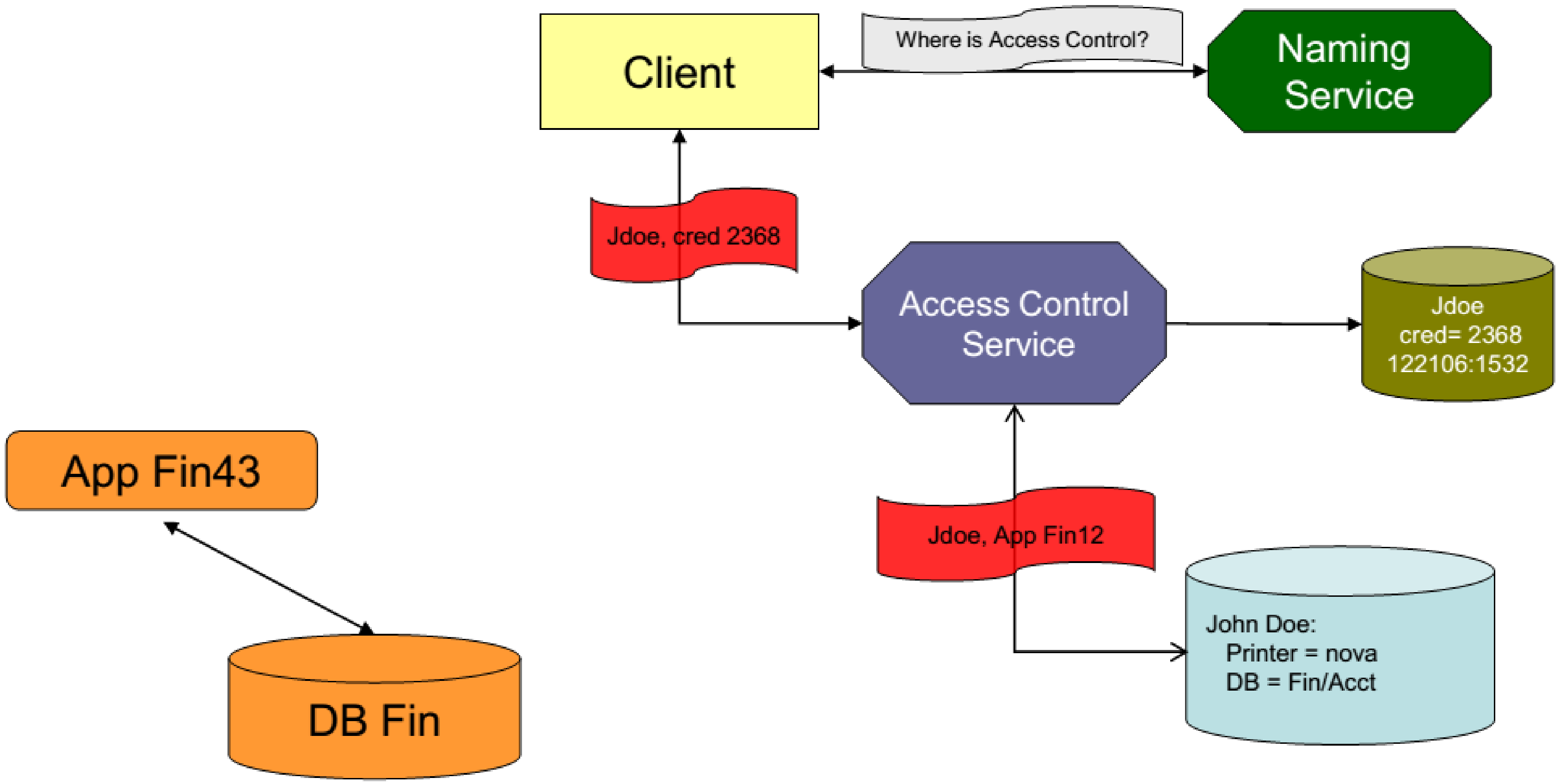
Security Services - Authentication



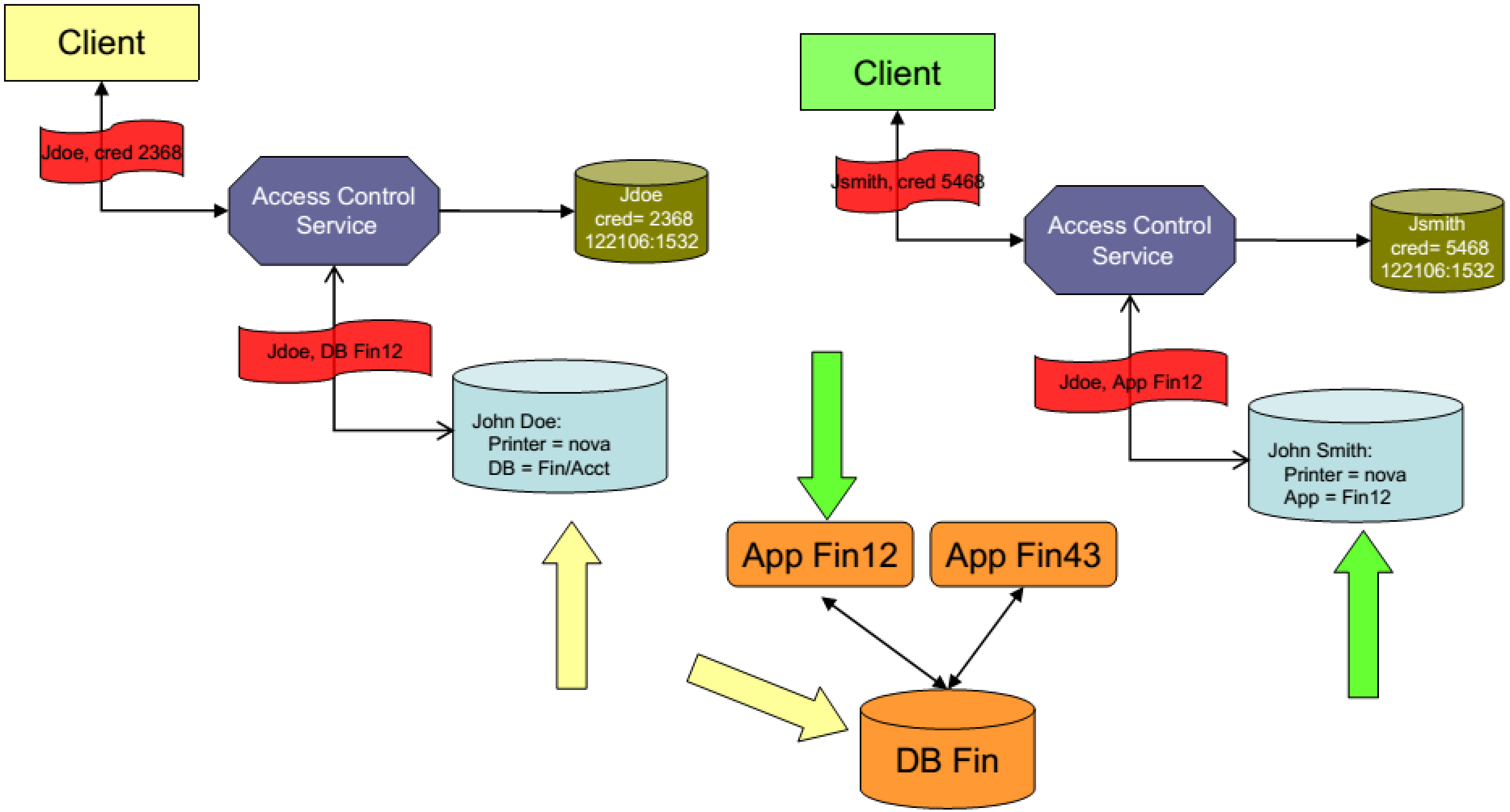
Integrated System Service: Authentication

- Different systems have different degrees of authentication
  - System X user logs on with user ID and password
  - System Y user must have a smart token
  - Systems have different levels of authentication and different types of authentication
- Establish a degree of trust between systems
  - *Why is it necessary to establish trust?*
    - Different systems have varying levels of security.
    - A system with weaker security (using only User + Password) cannot directly access a system with higher security (Smart Token, MFA, Biometric).
    - A common authentication mechanism is needed to ensure security during integration.

Security Services: Access Control



Security Services: Access Control



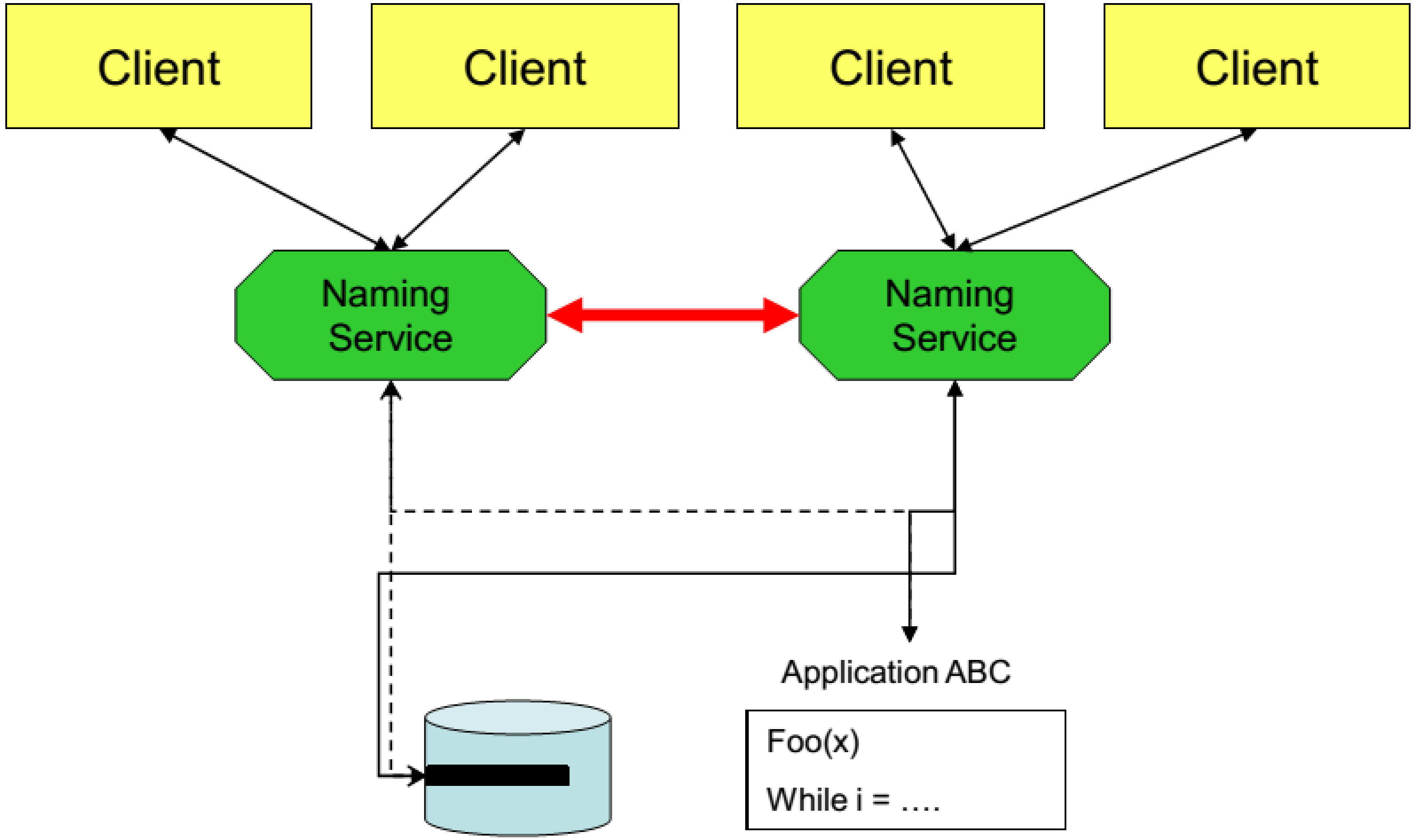
Integrated System Service: Access Control

- Policies conflict – example, users who have different roles in the organization can access data that should not be accessed
- Access to the data, but not the application
- Access to the application, but not the data
- Users are often assigned to groups
- Users can be in multiple groups
- Access determined by groups

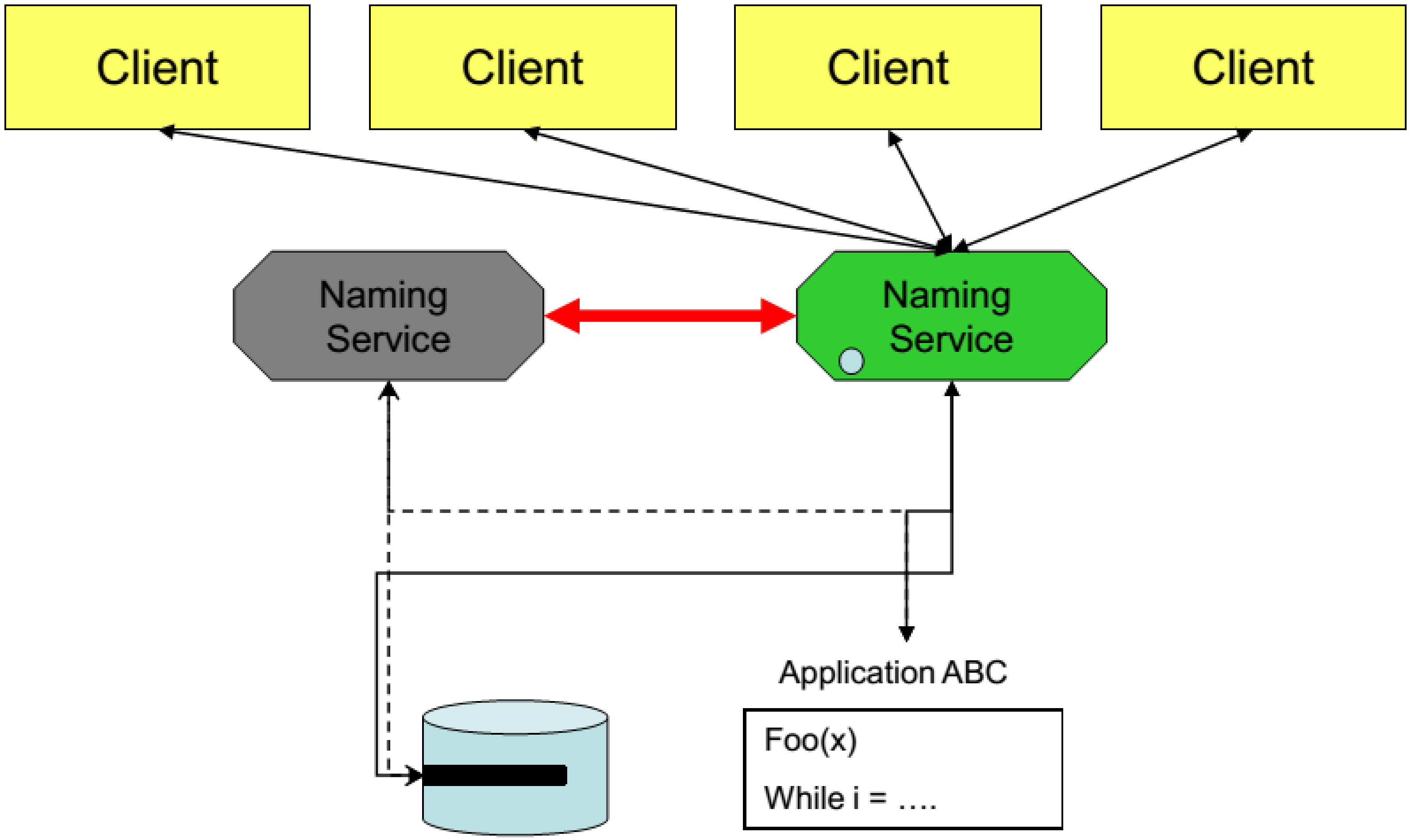
Global Properties of an Integrated System: Reliability

- Reliability – The ability of the system to provide necessary functionality despite unexpected events
  - Reliability primarily achieved with **replication** or **redundancy**
  - **Graceful degradation** of functionality when full functionality cannot be achieved
  - **Transparent to the user** when possible
  - Definition of **failure difficult to identify**

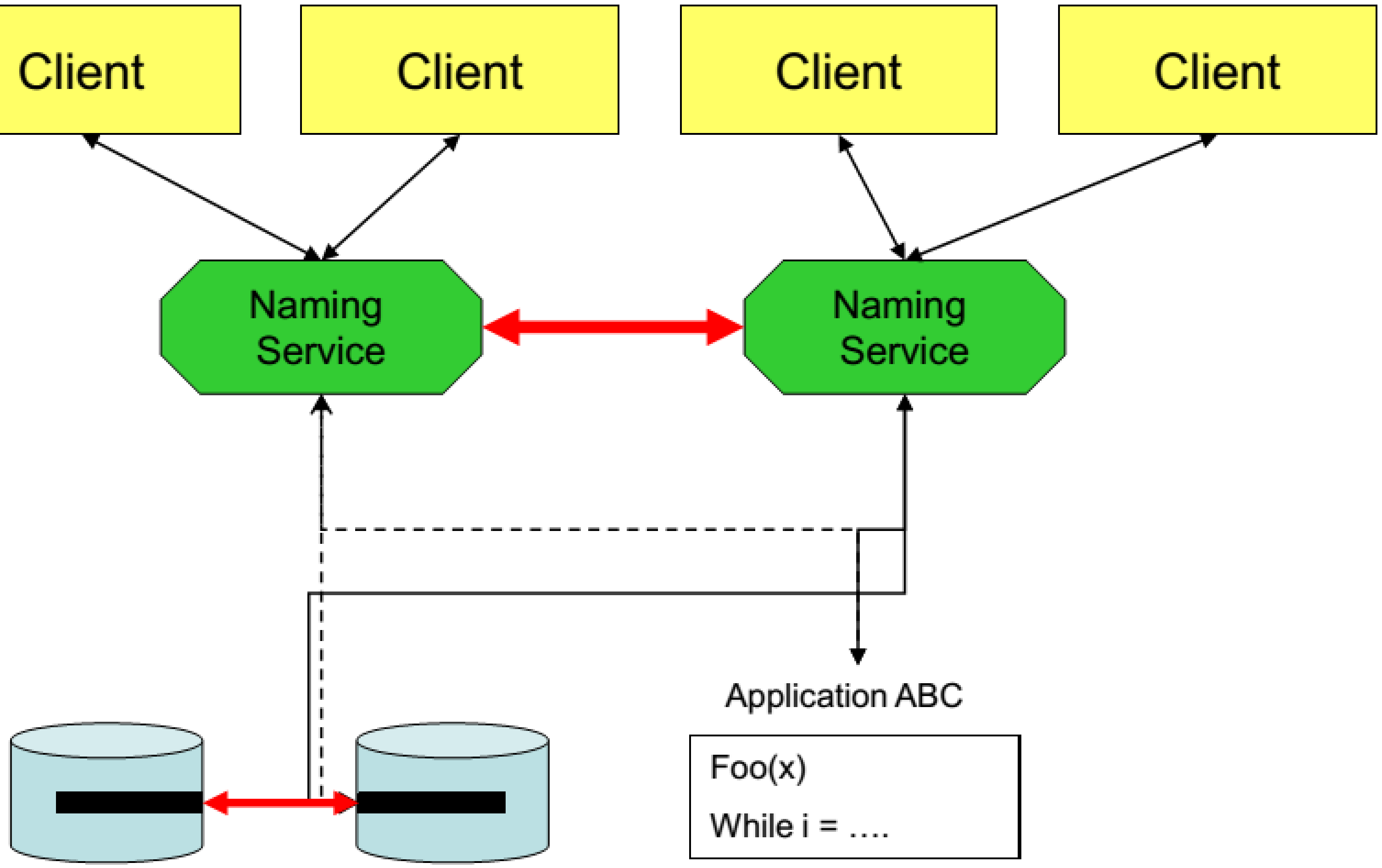
Reliability Services: Balancing



Reliability Services: Balancing



Reliability Services: Data Redundancy



Exercise – ERD Design for Access Control System

- **Scenario:** You are part of a team developing an internal management software for a large company. The system includes: Employee Management, Payroll, Reports, Document Management, and System Administration.
  To ensure security and proper authorization, the company requires a **Role-Based Access Control (RBAC) system**.

Business Requirements:

- Each user can have **one or more roles** (e.g., Employee, Manager, Accountant, Admin).
- Each role can have **one or more permissions** (e.g., view, add, edit, delete).
- Each permission applies to **one or more specific functions** (e.g., "View" permission applies to both Employee and Report functions).
- A function can be associated with **multiple permissions**.
- Each function belongs to **one module**, and each module contains **one or more functions**.
- The system must be **flexible and extensible** for future roles, permissions, and functions.

Goal: Design an **ERD** that captures the relationships among Users, Roles, Permissions, Functions, and Modules.

Exercise – ERD Design for Access Control System

Tasks:

- Identify **primary and secondary entities** in the system.
- Build an **ERD** showing relationships among **Users, Roles, Permissions, Functions, and Modules**.
- Ensure the diagram includes **primary keys, foreign keys, and junction tables** if needed.
- Write a **brief explanation** (under 1 page) of your ERD design.

Submission Requirements:

- **ERD file** as an image or PDF (created in SQL Server Diagram).
- **Short documentation** describing your design.

Summary

- Modern systems rely on different types of services that often must be integrated:
  - Naming service: Finding other services and objects in the system
  - Security services: Includes authentication, logging, and access control
  - Reliability services: ensuring a fully functional or partially functional system when one or more services is not available

Q/A?